

Observations

(multi-view images)

Task: Is there a TV in the room?



RGB-D Images



Memory Representation

(vantage-aware)



Position: Bedroom
Evidence: Partially detected



Position: Living room
Evidence: Fully detected



Position: Kitchen
Evidence: Partially detected

Supporting Views



Selective Decision

(evidence-sufficient)

No Support

v0

TV is not visible

No

Adapter

Low Support

v1 v3

TV is partially visible

Wait for more coverage
and additional views

? Abstain

High Support

v2

The entire TV is visible

Cross-view spatial
consistency

High score

Yes

Structured Memory



Claim: TV exists.



Support Set: v1, v2, v3



Support Score: 3



Estimate Position: (x, y, z)

Each claim retains supporting
views and source poses.